

Timothy Pham

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EDUCATION

University of Michigan | Ann Arbor, MI

Master's of Engineering in Space/Aerospace Systems Engineering | Expected May 2027

University of Michigan | Ann Arbor, MI

Bachelor of Science in Aerospace Engineering | May 2026

WORK EXPERIENCE

Navistar Defense | Madison Heights, MI

Systems Engineering Intern | May 2026- Present

- Validated structural integrity of tow bumpers and chassis components across 12 projects using Finite Element Analysis (FEA) in Siemens NX, ensuring designs met customer requirements
- Designed CAD fixtures for custom applications and integrated specialized electronics and hardware into center console systems across 4 projects
- Built an HTML-based document management system integrated with cloud infrastructure and custom dashboards, centralizing data for 500+ engineering projects, and developed a quote automation tool that cut entry time by 80%

DTE Energy | Detroit, MI

Power Distribution Engineering And Design Intern | June 2025- December 2025

- Engineered city electrical grid conversion projects making use of ESRI ArcMap and AutoCAD Electrical, generating detailed work packages and circuit diagrams impacting 100K+ customers
- Orchestrated and implemented new transformers for coverage wide deployment across Detroit, Michigan
- Collected field data and assisted with construction plan development utilizing ESRI

Program Management Co-op | January 2025-June 2025

- Supported design and commissioning of electrical systems across 15 concurrent infrastructure projects
- Authored work orders and project proposals that streamlined planning workflows and reduced approval cycle time for infrastructure upgrades

Hyundai Motor Group | Ann Arbor, MI

Vehicle Energy Efficiency Intern | May 2024- August 2024

- Utilized ETAS INCA to analyze vehicle performance data and calibrate fuel-economy parameters through CAN bus vehicle communication networks in latest ICE models
- Researched hybrid propulsion architectures and extended-range electric vehicle systems, synthesizing findings into technical reports delivered to Korean engineering teams

EXTRACURRICULAR PROJECTS

Whistler Wave Plasma Instrumentation | Los Alamos National Laboratory (SPACE 582 Sponsored Project)

Team Member | August 2026-Present

- Developing a mission concept for a LANL plasma-contactor technique that raises local plasma density $\sim 100\times$ to shrink the loop antenna needed for ~ 10 kHz whistler-wave generation from ~ 200 m to ~ 30 m
- Conducted trade studies across platform antenna configuration, propellant, and orbit, and building power and mass budgets for a hollow-cathode contactor drawing up to 1 A of ion current
- Advancing the concept toward a flight proposal to LANL by assessing whether reflected whistler pulses can be measured to detect small space debris, defining antenna and receiver requirements, and coordinating with LANL, UCLA, and electric propulsion experts

University of Michigan | Ann Arbor, MI

Undergraduate Research Assistant | September 2025- April 2026

- Analyzed geomagnetic disturbance effects on spacecraft RF communication links to support ongoing space weather monitoring research
- Conducted simulations in STK and MATLAB scripts to evaluate ground station visibility and link budget

SKILLS

Programming & Analysis: Python, MATLAB/Simulink, C++, VBA, C, MBSE

CAD & Simulation: CATIA, SolidWorks, AutoCAD, STK, Star-CCM+, XFOIL, Siemens NX

Hardware Systems: Arduino, Sensor Integration, CAN Bus Systems, PCB, ESRI ArcMap, ETAS INCA